

OSI reference model, TCP/IP and internet protocols ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Draw seven OSI layers with one function and one protocol/device each
- Map TCP/IP four layers onto OSI
- Define protocol, TCP, IP, HTTP, FTP, SMTP, DNS, URL

Protocol. A set of rules that sender and receiver follow so communication is understood (format, timing, error handling).

7 Application	HTTP, SMTP, FTP, DNS
6 Presentation	encryption, compression, translation
5 Session	start, manage, end sessions
4 Transport	TCP, UDP — ports, reliability
3 Network	IP addressing, routing
2 Data Link	frames, MAC, switches
1 Physical	cables, bits, hubs, NICs

ISO OSI 7-layer model — learn bottom to top: Please Do Not Throw Sausage Pizza Away (Physical → Application).

LEARN IT

Physical: bits on wire. Data link: frames, MAC, switch. Network: packets, IP, router. Transport: segments, TCP (reliable) / UDP (fast). Session: connections. Presentation: encrypt/compress. Application: user programs.

TCP/IP layer	OSI layers	Examples
Application	5+6+7	HTTP, SMTP, FTP, DNS
Transport	4	TCP, UDP
Internet	3	IP, ICMP
Network access / Link	1+2	Ethernet, Wi-Fi

Protocol	Use
TCP/IP	Suite that runs the internet
HTTP / HTTPS	Web pages (S = encrypted)
FTP	File transfer
SMTP	Sending email
POP3 / IMAP	Receiving email
DNS	Name → IP (www.biek.edu.pk)
URL	Uniform Resource Locator — web address

WORKED EXAMPLE / BOARD STYLE

URL example: <https://www.biek.edu.pk/index.html> — protocol https, host www.biek.edu.pk, path /index.html.

HOW BIEK ASKS THIS

2026 long: Draw and explain OSI. Also MCQ: seven layers; SMTP for email. Write one sentence per layer and a small stack diagram.

اردو میں سمجھیں

OSI ماڈل سات تہوں میں نیٹ ورک سمجھاتا ہے: فزیکل سے ایپلی کیشن تک۔ TCP/IP انٹرنیٹ کا عملی ماڈل ہے (چار تہیں)۔ پروٹوکول قواعد ہیں: SMTP ای میل بھیجنے، HTTP ویب، FTP فائل۔ URL ویب ایڈریس ہے۔

CHECK YOURSELF

1. How many OSI layers?
2. Which protocol sends email?

Answers: 1. Seven · 2. SMTP

MCQs

1. OSI model has:
 - a. 4 layers
 - b. 5 layers
 - c. 6 layers
 - d. 7 layers
2. URL stands for:
 - a. Universal Resource Locator
 - b. Uniform Resource Locator
 - c. Unique Resource Locator
 - d. Ultimate Resource Locator
3. HTTP works at the OSI:
 - a. Physical layer
 - b. Network layer
 - c. Application layer

d. Data link layer

Key: 1=d, 2=b, 3=c

Short questions (3 marks each)

1. Write the seven layers of the OSI model.
2. Differentiate TCP/IP and OSI.
3. What is a protocol? Give three examples.

Model short answers

1. Physical, Data Link, Network, Transport, Session, Presentation, Application — with one function each.
2. OSI is a 7-layer ISO reference model. TCP/IP is the 4-layer practical internet model. TCP/IP application covers OSI 5–7.
3. Rules of communication. TCP, IP, HTTP, SMTP, FTP.

Long questions (board Section C)

1. Draw and explain the OSI reference model. Give protocols at each layer.